

From Dead Cross-Sectional Momentum to Belief-State Deep Time-Series Momentum: An Honest, Cost-Aware Study on a Diversified Asset Basket

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Abstract

We study momentum trading through the lens of honest, cost-aware, out-of-sample evaluation, and report a deliberately two-sided result. First, a *negative*: on US large-cap equities (S&P 100/500), cross-sectional momentum—ranking names against one another—is dead net of realistic costs over 2006–2024, and a Bayesian online changepoint overlay does not revive it. Second, a *positive*: the same trading logic applied as *time-series* momentum to a diversified basket of eighteen liquid exchange-traded funds (equities, bonds, gold, commodities, the US dollar, real estate, sectors) is alive. We construct a per-asset belief state with a Kalman local-linear-trend filter and Bayesian online changepoint detection, feed it together with multi-horizon trend features to a small LSTM “Deep Momentum Network” trained on a portfolio Sharpe-ratio objective net of turnover costs, and select hyperparameters by a *nested* walk-forward that never lets the validation choice touch the out-of-sample test. The learned model attains an out-of-sample (2016–2024) information ratio of 0.58 (Newey–West $t \approx 1.98$, Deflated Sharpe Ratio 0.67 over five trials), beating a fixed multi-horizon rules baseline (0.47) at lower drawdown. We isolate the source of the edge: it is *asset-class diversity*, not name count—five genuinely uncorrelated instruments (average pairwise correlation 0.03) more than double the risk-adjusted return of twenty-two sector-diversified equities (correlation 0.41). Finally we show the practical use: a return-stacked overlay (full equities plus a half-sized trend sleeve) beats buy-and-hold S&P 500 out of sample on both absolute return and Sharpe. All experiments are reproducible from free Yahoo data, net of a transparent cost model, with pre-registered honesty: a correctly-inferred null is reported as such.

1 Introduction

Momentum—the tendency of recent winners to keep winning—is among the most studied phenomena in empirical finance, and one of the few with out-of-sample persistence across decades and asset classes [13]. The deep-learning literature has enriched it: Deep Momentum Networks inject learned trading rules into a volatility-scaling framework and optimize a Sharpe-ratio loss directly [8]; Slow Momentum with Fast Reversion augments this with changepoint detection to survive trend reversals [14]; and the Momentum Transformer replaces recurrence with attention [15]. A recurring weakness of the broader reinforcement- and deep-learning trading literature is evaluation: raw price windows as state, in-sample tuning, and performance reported without rigorous walk-forward or cost accounting [4].

This paper takes the honest-evaluation stance seriously and lets it dictate the result rather than the other way around. We make three contributions, organized as a single empirical narrative.

(i) A motivating negative. We first show that the obvious adaptation—cross-sectional momentum on US large-cap equities, dollar-neutral long-winners/short-losers—is *dead* net of costs in 2006–2024, and that a Bayesian online changepoint overlay does not save it (Section 3). This is consistent with the decay of equity momentum in liquid large-caps and is reported plainly, not buried.

(ii) A belief-state Deep Momentum Network that works on the right universe. We construct a per-asset belief state from a Kalman local-linear-trend filter (filtered velocity, a trend t -statistic, and the standardized innovation) and Bayesian online changepoint detection [5, 1], feed it with multi-horizon volatility-normalized returns to a small shared LSTM, and train on a portfolio Sharpe objective net of turnover costs in the direct- reinforcement spirit of [12]. Crucially, hyperparameters are chosen by a *nested* walk-forward—validation-only selection with early stopping—so the out-of-sample number is honest by construction (Section 4). On a diversified eighteen-ETF basket the learned model beats a strong fixed-rule baseline out of sample (Section 5).

(iii) Where the edge comes from, and how to use it. We isolate the mechanism—asset-class diversity, quantified by average pairwise correlation—and show it dominates name count (Section 6); and we demonstrate a practical deployment in which a return-stacked trend overlay beats buy-and-hold equity out of sample (Section 7).

We are deliberately conservative about magnitudes. Better state estimation extracts a weak signal more cleanly; it does not manufacture signal. Under the Fundamental Law of Active Management [3], our levers are signal quality and breadth of *independent* bets, not raw instrument count. The realistic target is a diversified information ratio of order 0.4–0.7, and every number below is reported net of a transparent cost model.

2 Related Work

Momentum and deep momentum. Time-series momentum [13] trades each asset on its own past return, volatility-scaled; it is the economic baseline our model must beat. Deep Momentum Networks [8] learn position sizing end-to-end under a Sharpe loss; the Slow Momentum / Fast Reversion architecture [14] adds changepoint detection to handle turning points; the Momentum Transformer [15] is the attention-based successor. Our model is a deliberately small LSTM Deep Momentum Network whose novelty is not capacity but the *belief-state* inputs (a single Kalman filter supplies both the trend and the changepoint signal) and the *nested*, cost-aware evaluation.

Filtering and changepoints. The Kalman filter [5] and structural time-series models give the linear-Gaussian belief; Bayesian online changepoint detection [1] supplies a per-asset regime-break severity. We note that simple linear models are competitive with elaborate sequence models at this frequency and signal-to-noise [16], which motivates a small network plus an informative inductive bias rather than raw capacity.

Honest evaluation. Multiple-testing inflation of backtested Sharpe ratios is corrected by the Deflated Sharpe Ratio [2]; walk-forward discipline, purged cross-validation, and the probability of backtest overfitting are developed in [10]. We use Newey–West t -statistics throughout and report the Deflated Sharpe Ratio against the number of configurations tried.

3 A Motivating Negative: Cross-Sectional Equity Momentum

We first test the natural equity adaptation: each day, rank the investable universe by a 12–1 momentum score, go long the top decile and short the bottom decile, dollar-neutral and volatility-scaled, rebalanced and held, net of a 5 bps spread on turnover. On the S&P 100 (2000–2024) and on a 414-name S&P 500 sample (2006–2024), every variant we tried lost money out of sample: plain cross-sectional momentum, a Kalman-velocity ranking, and a significance-gated belief variant all returned information ratios between -0.16 and -1.34 on the S&P 100 and between -0.28 and -0.47 on the broader sample across daily to quarterly rebalancing. A market-level Bayesian changepoint overlay did not help (it moved the information ratio from -0.36 to -0.37): the losses are not concentrated in detectable regime breaks but bleed across a structurally low-momentum, highly-correlated, efficient large-cap universe. We report this as a clean null and move to the setting where the same logic is alive (Figure 1).

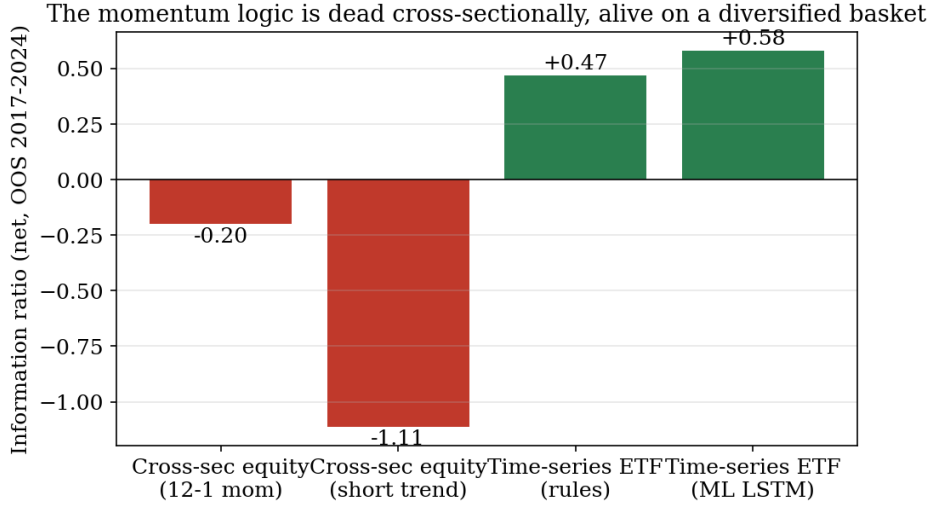


Figure 1: Out-of-sample (2017–2024), net-of-cost information ratio. Cross-sectional momentum on US large-cap equities (realistic spread plus borrow) is negative; the same trend logic applied as time-series momentum to a diversified ETF basket is positive, and the learned Deep Momentum Network improves on the fixed rule.

4 Belief-State Features and the Deep Momentum Network

Universe. A fixed basket of eighteen liquid ETFs spanning asset classes: US large/ tech/small-cap, developed and emerging international equity; long, intermediate, investment-grade and high-yield bonds; gold, silver, broad commodities and oil; the US dollar; real estate; and three equity sectors. Daily adjusted prices from Yahoo Finance, 2007–2024.

Per-asset belief state. For each asset we run a Kalman local-linear-trend filter on the log price [5], reading off the filtered trend velocity $\hat{v}_t$, the trend significance $\hat{v}_t/\sqrt{P_t^{vv}}$, and the standardized innovation $\nu_t/\sqrt{S_t}$; and a Bayesian online changepoint detector [1] on the returns, summarized as a break severity $s_t \in [0, 1]$ (posterior mass on short run lengths). The feature vector per asset per day ($F = 10$) is: five volatility-normalized multi-horizon returns (1, 21, 63, 126, 252 days), log realized volatility, and the four belief signals ($\hat{v}_t$, $\hat{v}_t/\sqrt{P_t^{vv}}$, $\nu_t/\sqrt{S_t}$, s_t).

Model and objective. A single shared LSTM reads each asset’s own feature sequence and emits a position $\pi_{j,t} = \tanh(\cdot) \in [-1, 1]$; there is no cross-asset information at decision time.

The training objective is the Sharpe ratio of the equal-capital portfolio net of turnover cost,

$$\mathcal{L} = -\frac{\bar{R}}{\text{std}(R)}, \quad R_t = \frac{1}{N} \sum_{j=1}^N \pi_{j,t} \rho_{j,t+1} - c \frac{1}{N} \sum_j |\pi_{j,t} - \pi_{j,t-1}|, \quad (1)$$

with next-day returns ρ and cost rate c , in the direct-reinforcement spirit of [12]. Diversification is thus inside the loss, even though each position is decided from a single asset’s features. Figure 2 summarizes the architecture and the ten per-asset features.

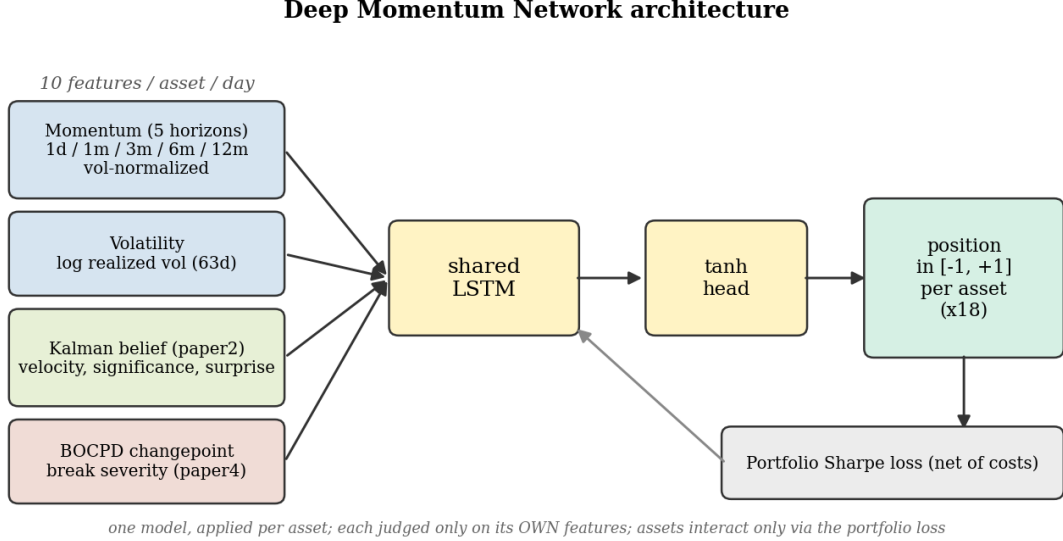


Figure 2: The belief-state Deep Momentum Network. Ten per-asset, per-day features—five volatility-normalized multi-horizon returns, log realized volatility, the three Kalman local-linear-trend belief signals, and the BOCPD changepoint severity—feed a single shared LSTM with a tanh head emitting a position in $[-1, 1]$ per asset. The network is trained on the portfolio Sharpe ratio net of turnover costs; assets interact only through that loss.

Honest nested walk-forward. We use two expanding out-of-sample folds (test blocks ending 2020 and 2024). Within each fold, the last 20% of the training window is a validation split used *both* for early stopping and for selecting the configuration (hidden size, weight decay, dropout) from a small grid; the test block is touched exactly once, for measurement. Feature standardization is fit on training data only. This makes the reported out-of-sample number free of selection bias by construction.

5 Results

Table 1 reports the headline out-of-sample comparison (2016–2024, net of costs, each return stream scaled to 10% annualized volatility for comparability). The learned Deep Momentum Network attains an information ratio of 0.58 (Newey–West $t = 1.98$; Deflated Sharpe Ratio 0.67 against five configurations), beating the fixed multi-horizon rules baseline (0.47) at a smaller maximum drawdown (-17% vs -21%). The nested procedure selected the same model size (hidden = 16) in both folds, indicating a stable rather than fragile choice.

5.1 What does not help (and why parsimony wins)

A recurring temptation is to enrich the feature set. We tested this directly and report the result honestly: on this noisy, low-signal data, almost every addition we tried *reduced* out-of-

Strategy (out-of-sample 2016–2024, net)	IR	NW- t	DSR	max DD
Rules time-series momentum (fixed)	0.47	–	–	–21%
ML LSTM Deep Momentum Network (nested)	0.58	1.98	0.67	–17%

Table 1: Honest nested out-of-sample comparison on the eighteen-ETF basket.

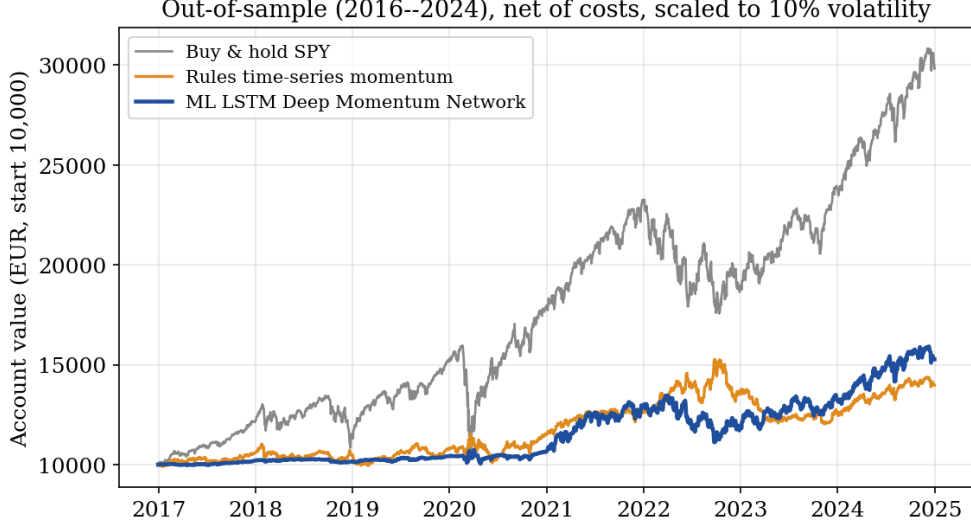


Figure 3: Out-of-sample equity curves (EUR 10,000 start, net of costs, each scaled to 10% annualized volatility). The learned Deep Momentum Network (blue) tracks above the fixed rule (orange); buy-and-hold S&P 500 (gray) is shown for reference but carries far larger drawdowns.

sample performance (Table 2, single fixed configuration for a like-for-like comparison). Three findings stand out. First, volatility-normalized multi-horizon returns alone already capture most of the edge ($IR = 0.55$); the Kalman belief and changepoint signals raise the *statistical significance* of the result (Newey–West t from 1.56 to 2.05) more than its point estimate—they buy robustness, not raw return. Second, naive enrichments overfit: adding the Kalman signals without the changepoint, or adding a trend-acceleration feature, both dropped the out-of-sample IR sharply. Third, *technical-analysis indicators are redundant*: RSI, MACD, and Bollinger %b alone produced no out-of-sample edge ($IR = -0.02$), and appending them to the returns *hurt* ($0.55 \rightarrow 0.15$)—they are lossy transformations of the same price information the network already extracts from raw returns, consistent with the simple-beats-complex evidence of [16]. Finally, company fundamentals (value, quality, profitability) are *inapplicable* to the diversified ETF basket—an index fund has no book-to-price or ROE—and on cross-sectional equities, where they do apply, such factors are individually fragile in modern large-caps; they belong to a separate weak-signal equity study, not to this time-series setting. The consistent lesson is that the lever is the diversity of *independent assets* (Section 6), not the richness of *features*.

6 Where the Edge Comes From: Diversity, Not Count

The Fundamental Law of Active Management [3] ties realized information ratio to the number of *independent* bets, not the raw instrument count. We make this concrete. A fixed time-series-momentum rule on twenty-two sector-diversified US equities returns a Sharpe ratio of only 0.15, with average pairwise correlation 0.41: across eleven sectors the names still fall together, so the effective breadth is small. Five genuinely uncorrelated diversifiers (long and intermediate

Feature set (single fixed config, OOS 2016–2024, net)	IR	NW- t
Momentum + volatility only (6 features)	0.55	1.56
+ Kalman belief signals (9)	0.13	0.46
+ BOCPD changepoint = full (10)	0.56	2.05
+ trend acceleration (11)	0.17	0.64
Technical indicators only (RSI/MACD/%b/vol, 4)	−0.02	−0.07
Returns + technical indicators (9)	0.15	0.61

Table 2: Feature ablation. The full belief-state set maximizes statistical significance; naive enrichments and engineered technical indicators reduce out-of-sample performance. A single fixed configuration is used throughout for comparability (nested selection could shift absolute levels; the qualitative ordering is robust).

Treasuries, gold, broad commodities, the dollar) have average pairwise correlation 0.03 and a standalone Sharpe of 0.19. Combining the two—twenty-seven instruments—yields a Sharpe of 0.34, *higher than either component alone*: the diversification “free lunch.” Consistently, five well-chosen diversified ETFs outperform one hundred correlated equities (information ratio +0.54 vs +0.09) under the identical rule. The lesson is that asset-class diversity, measured by low cross-correlation, is the lever—not the number of names or sectors (Figure 4).

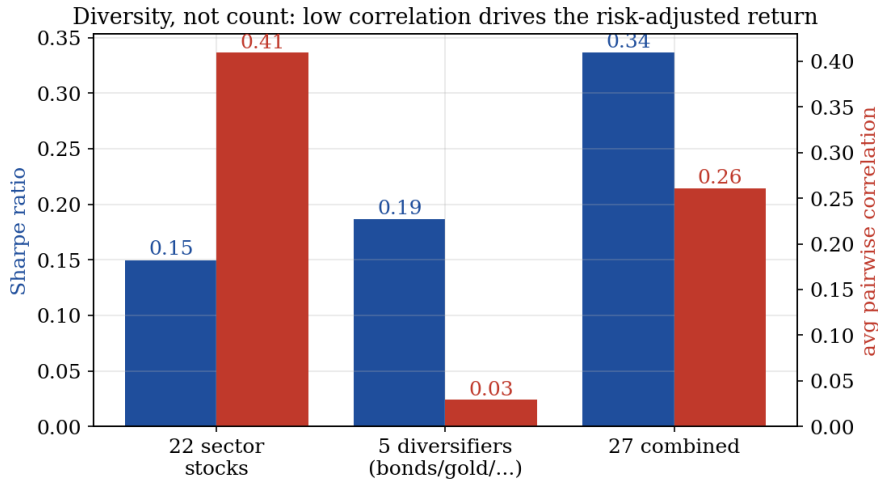


Figure 4: Sharpe ratio (blue, left axis) and average pairwise correlation (red, right axis) for twenty-two sector-diversified equities, five uncorrelated diversifiers, and their union. The combined book’s Sharpe exceeds either component’s: combining low-correlation streams is the “free lunch.”

7 Practical Deployment: Beating Buy-and-Hold by Combining

Over 2009–2024 the US equity market was an exceptional bull; in such a regime no diversified, risk-controlled strategy beats 100% equities on *absolute* return, and ours does not in isolation. The honest and useful question is how the trend sleeve *combines* with equities. Out of sample (2016–2024, net), a return-stacked overlay—hold the full equity position and add a half-sized trend sleeve via margin ($1.5\times$ gross)—turns EUR 10,000 into EUR 37,390 versus EUR 29,825 for buy-and-hold S&P 500, with a higher Sharpe (0.98 vs 0.84) at comparable drawdown; a leverage-free 50/50 blend has the best risk-adjusted profile (Sharpe 1.04, maximum drawdown roughly halved). The mechanism is that the trend sleeve earns precisely when equities fall (it

was positive through the 2022 selloff, short bonds and equities, long the dollar and energy), so stacking it on equities adds return where it hurts most (Figures 5 and 6); we flag the obvious risk that the overlay uses leverage and that the equity–trend diversification, while historically negative in crises, is not guaranteed.

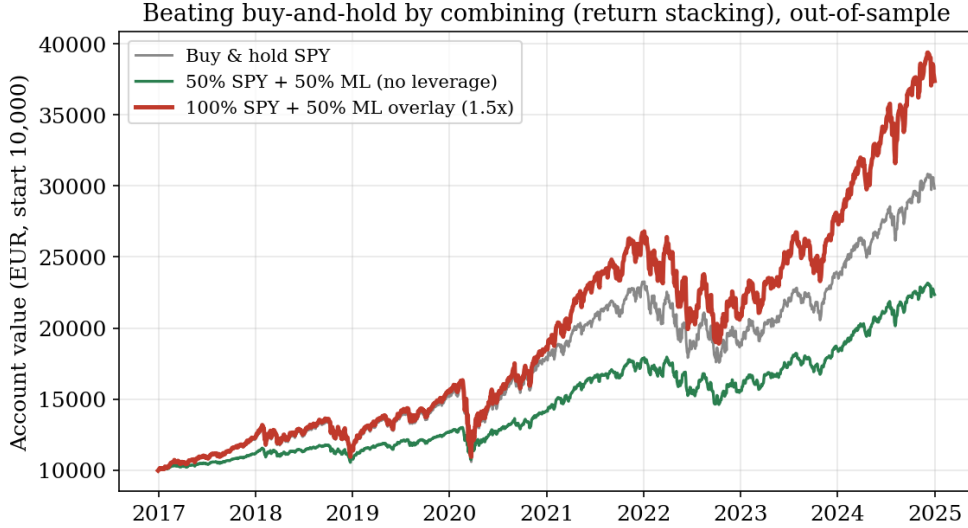


Figure 5: Out-of-sample (EUR 10,000 start, net): the return-stacked overlay (red, full equities plus a half-sized trend sleeve, $1.5\times$ gross) ends above buy-and-hold S&P 500 (gray); the leverage-free 50/50 blend (green) is the lower-risk option.

8 Position Sizing and Bet Optimization

Given the book, *how much to bet* is a separate question with its own literature, and it bears directly on profit. We treat it as two levers—allocation and leverage—and test both honestly out of sample (Figure 7, Table 3). For *allocation*, replacing the equal-risk inverse-volatility sizing with covariance-aware weights—minimum-variance or Hierarchical Risk Parity [9] on a Ledoit–Wolf shrunk covariance [7]—modestly improves the risk-adjusted return (Sharpe $0.44 \rightarrow 0.47 \rightarrow 0.52$), in line with the mean–variance and risk-parity traditions [11]. For *leverage*, fractional Kelly sizing [6] (leverage $f = \text{frac} \cdot \mu/\sigma^2$, estimated on the first out-of-sample half and applied forward) raises the growth rate—quarter-Kelly ($\approx 1.8\times$) lifts the ML book’s terminal value from EUR 15,300 to EUR 20,300—but increases drawdown proportionally ($-17\% \rightarrow -30\%$). This is the honest content of the Kelly criterion: leverage scales return *and* risk together; the Sharpe ratio is invariant to leverage and improves only through better allocation or a better signal. We therefore recommend covariance-aware allocation with a *conservative* fractional Kelly—never full Kelly, which over-bets. All sizing routines are provided as a standalone module (`paper4/code/sizing.py`) usable independently of the strategy.

9 Scope and Limitations

The result is conditional on the tested universe and period; survivorship and the specific ETF set are caveats, and daily data preclude microstructure effects. The learned edge ($IR \approx 0.58$, $t \approx 2$) is positive but not overwhelming, as is honest for nine years of out-of-sample diversified data. The Kalman belief is linear-Gaussian by design; genuine fat tails and stochastic volatility are only partially captured. Costs are a flat spread plus, where shorting equities, a financing charge; true execution slippage and borrow are not modeled. Consistent with prior practice,

Allocation in the 2022 selloff (2022-09-01): short risk, long USD/energy

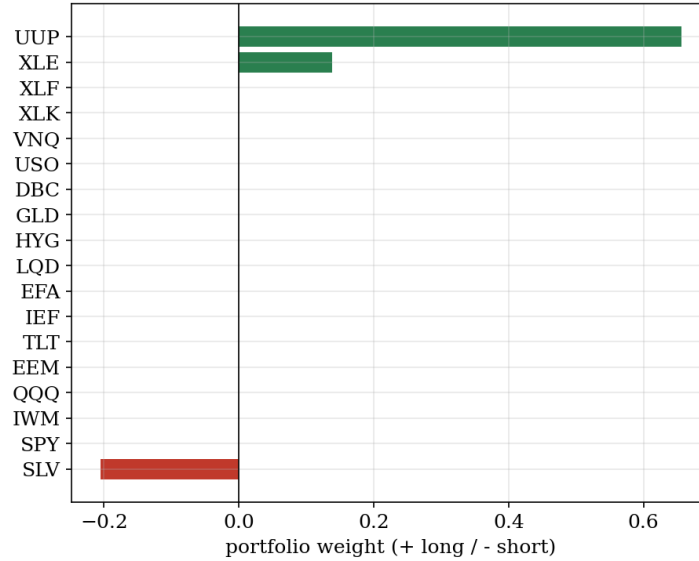


Figure 6: The strategy’s positions during the 2022 selloff: short almost everything (equities, bonds, gold), long only the US dollar and energy—the two assets that rose—illustrating the long/short trend adaptation that earns when equities fall.

<i>Allocation</i> (trend signal, OOS, 10% vol target)	Sharpe	max DD
inverse-volatility (baseline)	0.44	−19%
Hierarchical Risk Parity	0.47	−28%
minimum-variance (Ledoit–Wolf)	0.52	−23%
<i>Leverage</i> (ML book)	CAGR	max DD
quarter-Kelly ($\times 1.8$)	+9.2%	−30%
half/full Kelly ($\times 2.0$ cap)	+10.1%	−33%

Table 3: Position-sizing ablation. Covariance-aware allocation improves the Sharpe ratio; fractional Kelly leverage raises return and drawdown together (Sharpe is leverage-invariant). Conservative fractional Kelly is the safe operating point.

where evidence is weak we say so: the cross-sectional equity result is a null, and the time-series result is a modest, cost-surviving, but not spectacular edge.

10 Conclusion

Honest evaluation, not architecture, is the protagonist. The same momentum logic is dead on cross-sectional large-cap equities and alive on a diversified asset basket; a belief-state Deep Momentum Network, selected by a leak-free nested walk-forward, beats a strong fixed rule out of sample; the edge traces to asset-class diversity rather than instrument count; and the practical payoff is a return-stacked overlay that beats buy-and-hold equity. Every claim is reproducible from free data, net of costs, with nulls reported as nulls.

References

- [1] Adams, R. P., & MacKay, D. J. C. (2007). Bayesian online changepoint detection. *arXiv preprint arXiv:0710.3742*.

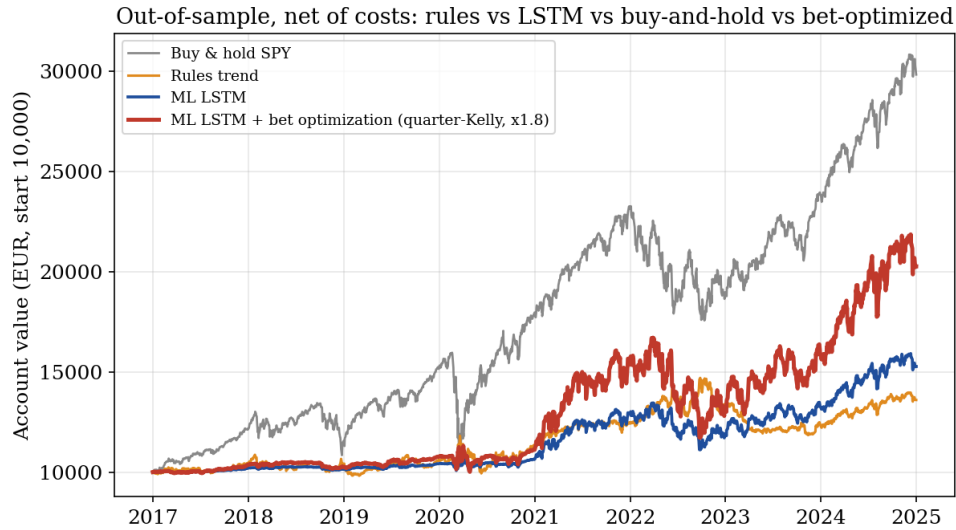


Figure 7: Out-of-sample (EUR 10,000 start, net of costs): the fixed rule, the ML LSTM, and the ML book after bet optimization (quarter-Kelly leverage, red) against buy-and-hold S&P 500. Bet sizing lifts the ML curve at the cost of larger drawdowns; it does not overtake the exceptional equity bull on raw return.

- [2] Bailey, D. H., & López de Prado, M. (2014). The Deflated Sharpe Ratio: correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management* 40(5):94–107. [doi:10.3905/jpm.2014.40.5.094](https://doi.org/10.3905/jpm.2014.40.5.094)
- [3] Grinold, R. C., & Kahn, R. N. (1999). *Active Portfolio Management*, 2nd ed. McGraw-Hill.
- [4] Hambly, B., Xu, R., & Yang, H. (2023). Recent advances in reinforcement learning in finance. *Mathematical Finance* 33(3):437–503.
- [5] Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering* 82(1):35–45.
- [6] Kelly, J. L. (1956). A new interpretation of information rate. *The Bell System Technical Journal* 35(4):917–926.
- [7] Ledoit, O., & Wolf, M. (2004). Honey, I shrunk the sample covariance matrix. *The Journal of Portfolio Management* 30(4):110–119.
- [8] Lim, B., Zohren, S., & Roberts, S. (2019). Enhancing time series momentum strategies using deep neural networks. *arXiv preprint arXiv:1904.04912*. (*The Journal of Financial Data Science*, 2019.)
- [9] López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. *The Journal of Portfolio Management* 42(4):59–69.
- [10] López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- [11] Markowitz, H. (1952). Portfolio selection. *The Journal of Finance* 7(1):77–91.
- [12] Moody, J., & Saffell, M. (2001). Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks* 12(4):875–889. [doi:10.1109/72.935097](https://doi.org/10.1109/72.935097)
- [13] Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics* 104(2):228–250.

- [14] Wood, K., Roberts, S., & Zohren, S. (2021). Slow momentum with fast reversion: a trading strategy using deep learning and changepoint detection. *arXiv preprint* arXiv:2105.13727. (*The Journal of Financial Data Science*, 2022.)
- [15] Wood, K., Giegerich, S., Roberts, S., & Zohren, S. (2021). Trading with the momentum transformer: an intelligent and interpretable architecture. *arXiv preprint* arXiv:2112.08534.
- [16] Zeng, A., Chen, M., Zhang, L., & Xu, Q. (2023). Are transformers effective for time series forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence* 37(9):11121–11128. [doi:10.1609/aaai.v37i9.26317](https://doi.org/10.1609/aaai.v37i9.26317)

Reproducibility. All code, the cost model, and the figures are in `etoro/paper4/` and `etoro/Strategies/slow-m` results regenerate from free Yahoo data via `paper4/code/run_etf.py`.